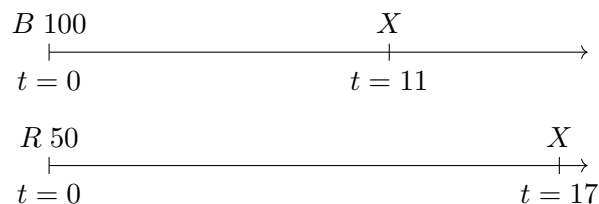


Problem 8.13

Bruce and Robbie each open up new bank accounts at time 0. Bruce deposits 100 into his bank account, and Robbie deposits 50 into his. Each account earns an annual effective discount rate of d . The amount of interest earned in Bruce's account during the 11th year is equal to X . The amount of interest earned in Robbie's account during the 17th year is also equal to X . Calculate X .

Solution

We represent the deposits and the interest earned in the corresponding year:



If the annual effective discount rate is d , then the annual accumulation factor is

$$\frac{1}{1-d}.$$

The interest earned during year n on an initial deposit P is the accumulated value at the end of that year multiplied by the discount rate:

$$P \left(\frac{1}{1-d} \right)^n d.$$

Therefore,

$$100 \left(\frac{1}{1-d} \right)^{11} d = X, \quad 50 \left(\frac{1}{1-d} \right)^{17} d = X.$$

Equating both expressions:

$$\begin{aligned} 100 \left(\frac{1}{1-d} \right)^{11} d &= 50 \left(\frac{1}{1-d} \right)^{17} d, \\ 2 &= \left(\frac{1}{1-d} \right)^6. \end{aligned}$$

Then

$$\begin{aligned}2^{1/6} &= \frac{1}{1-d}, \\2^{1/6} - 2^{1/6}d &= 1, \\2^{1/6} - 1 &= 2^{1/6}d, \\d &= 1 - 2^{-1/6}.\end{aligned}$$

Finally,

$$\begin{aligned}X &= 100 \left(\frac{1}{1-d} \right)^{11} d \\&= 100 \left(2^{1/6} \right)^{11} \left(1 - 2^{-1/6} \right) \\&\approx 38.8972768.\end{aligned}$$